

Jical Patel

jicalharish15102002@gmail.com | +1 (216) 856-2902 | linkedin.com/in/jical-patel | Portfolio | Ohio, USA

Professional Summary

Software Developer with experience in **Android development**, **web applications**, **machine learning**, **data analysis**, **Agile delivery**, and **technical support**. Skilled in **Java**, **Python**, **Kotlin**, **SQL**, **REST APIs**, **Firebase**, **Django**, **Next.js**, and **practical software engineering**. Built solutions across **healthcare AI**, **audio ML**, **GPU systems**, **recommendation analytics**, **blockchain apps**, and **mobile platforms**.

Technical Skills

- **Programming:** Java, Kotlin, Python, C, C++, SQL, Data Structures and Algorithms
- **Mobile Development:** Android Studio, Firebase, REST APIs, XML, Jetpack Compose
- **Web / Backend:** Next.js, Django, MongoDB, MySQL, SQLite, Cloud Firestore
- **Machine Learning / Data:** TensorFlow, Keras, OpenVINO, FastSpeech2, HiFi-GAN, Gradio, Tableau, Statistical Testing
- **Tools / Practices:** Git, GitHub Workflows, Debugging, Agile Development, UI/UX Design, Performance Optimization

Education

Case Western Reserve University, Ohio, USA Aug 2024 – May 2026
Master's in Computer Science

- Focused on **software engineering**, **systems**, **data science**, and **applied machine learning**.

San Francisco State University, California, USA Jul 2023 – Aug 2023
Data Analysis with Computer Application

- Earned an **A grade** after completing scholarship-supported coursework in **statistical testing** and applied data analysis.

LDRP-ITR, Gujarat, India Sep 2020 – May 2024
Bachelor of Engineering in Computer Engineering

- Completed core coursework in **programming**, **databases**, **algorithms**, **mobile development**, and software project design.

Experience

Office Assistant – Siegal Lifelong Learning, Case Western Reserve University May 2025 – May 2026

- Assisted **200+ instructors** with classroom technology setup, troubleshooting, and program coordination across lifelong learning sessions.
- Computerized recurring **back-office tasks**, streamlined record handling, and strengthened communication with instructors, staff, and participants.

Software Developer – xLab / Progressive Insurance, Case Western Reserve University Aug 2025 – Dec 2025

- Delivered **1 Agile software application** using **GitHub workflows**, **iOS**, **Next.js**, **MongoDB**, and **Java**.
- Translated business requirements into functional product features while supporting sprint planning, testing, and team-based implementation.

Graduate Assistant – Case Western Reserve University Jan 2025 – Dec 2025

- Mentored **80+ students** in **Computer Security** through concept review, problem-solving sessions, and technical guidance.
- Graded **100+ assignments and exam papers** using course rubrics while providing clear academic feedback and coding support.

Android Developer – TOPS Technology Nov 2023 – Apr 2024

- Built Android application features using **Java**, **Kotlin**, and **Firebase** for authentication, storage, and app workflows.
- Created reusable **XML** and **Jetpack Compose** interfaces across core mobile screens to improve usability and consistency.
- Connected mobile apps with **REST APIs** to support real-time data exchange and reliable application functionality.

Projects

Singing Voice Synthesis Using Machine Learning Spring 2026

- Built an end-to-end **audio ML pipeline** converting lyrics into sung audio with **FastSpeech2 Conformer**, **HiFi-GAN**, transfer learning, CSD dataset analysis, and a **Gradio** interface.

Distributed HIP DNN Pipeline Fall 2025

- Engineered a **2-stage divided GPU pipeline** where **MPI producers** streamed sharded batches to **AMD HIP consumers** for DNN kernel execution.
- Benchmarked workload behavior across **throughput**, **latency**, and **scalability** to evaluate allocated GPU training performance.

MedCare Healthcare Recommendation Platform Jan 2025 – May 2025

- Built a **machine learning-based healthcare platform** that analyzed uploaded medical reports and generated disease predictions with up to **80% response accuracy**.
- Developed a **report-driven recommendation system** to deliver personalized health insights and improve accessibility of medical decision support.

StegAnagram: Image Steganography with VAEs and Visual Anagrams Spring 2025

- Built a **2-method image steganography system** using **Variational Autoencoders**, **Visual Anagrams**, diffusion models, and latent-space embedding.
- Evaluated **7 image-hiding experiments** across **4 metrics**, achieving VAE results of **0.019 detectability**, **52.3 MSE**, **31.7 PSNR**, and **0.63 SSIM**.

Co-Purchasing Analysis and Recommendation System	Fall 2024
Built an SQL-based market basket analysis workflow to identify product relationships, buying patterns, and recommendation opportunities.	
Uber Clone Ride-Hailing App	Nov 2023 – Feb 2024
Enhanced a blockchain-based ride-hailing system using smart contracts, Ethereum Rinkeby Testnet, crypto payments, GPS tracking, and automated fare calculation.	
Deep Learning Skin Condition Classifier	May 2023 – Aug 2023
Trained an AI classification model with 75% accuracy using TensorFlow, Keras, and OpenVINO, then deployed a mobile-friendly diagnosis interface.	

Research

Battery Efficiency through Thermal Analysis	Sep 2021 – Nov 2021
Published 1 research paper on battery efficiency optimization using thermal analysis in the International Journal of Research Publication and Reviews.	

Certifications

Developing Secure Software (LFD121) – Linux Foundation	Aug 2024 – Nov 2024
NPTEL Courses – Programming in Java and Python for Data Science	Mar 2023 – Apr 2023
State-level Workshop on Web Development using PHP Laravel	Mar 2021 – Apr 2021